Generation Conscious Laundry Dispenser Project

1. System Architecture

1.1 Hardware Platform

1.1.1 Raspberry Pi Compute Module 4 (CM4)

- The **Raspberry Pi CM4** serves as the core processing unit for the application. It offers robust performance with various configurations of RAM and eMMC storage, suitable for handling complex computational tasks and ensuring smooth operation.
- **Key Features**:
  - **Processor**: Quad-core ARM Cortex-A72 (64-bit) @ 1.5GHz.
  - **Memory**: 4GB
  - **Storage**: 8GB – 32GB
  - **Connectivity**: Support for Ethernet, USB, PCIe, and other interfaces.

1.1.2 Official Raspberry Pi IO Board

- The application utilizes the **official Raspberry Pi IO Board** as a carrier board for the CM4 module, providing necessary interfaces and connectors to integrate various peripherals seamlessly.
- **Functionality**:
  - **LCD Panel Connection**: The IO Board facilitates the connection of an LCD panel through dedicated **DSI (Display Serial Interface)** ports, enabling display capabilities for the application's graphical user interface.
  - **Peripheral Integration**: Provides multiple interfaces such as **HDMI, USB, Ethernet, DSI, GPIO pins**, and **serial ports**, allowing easy integration of additional peripherals like sensors, input devices, and custom hardware modules.
  - **Power Management**: Offers reliable power delivery and management to the CM4 and connected peripherals, ensuring stable and efficient operation.

1.1.3 LCD Panel

- The application employs an **LCD panel** connected via the IO Board's DSI interface to provide a user-friendly and interactive **Graphical User Interface (GUI)**.
- **Specifications**:
  - **Size**: 10.1” or 8”
  - **Resolution**: 1280 x 800
  - **Touch Capability**: Supports **single touch** inputs for intuitive user interactions.
- **Integration**:
  - The LCD panel is configured and controlled through **PyQt 5** within the application, delivering responsive and visually appealing interfaces.
  - Proper calibration and driver setup ensure accurate touch response and display performance.

1.1.4 Custom Kiosk Motor Control PCB

- A **custom Kiosk Motor Control PCB** is integrated to manage and control motorized components within the kiosk system.
- **Communication**:
  - Utilizes **serial communication** facilitated by the **pySerial 3.5** library to send commands and receive feedback from the motor control PCB.
  - Ensures precise and reliable control over mechanical operations, contributing to the system's overall functionality and user experience.

1.2 Software Environment

1.2.1 Operating System

- **Raspberry Pi OS Bookworm (64-bit)**: The system runs on the latest **Raspberry Pi OS Bookworm 64-bit**, providing a stable and secure platform optimized for the ARM architecture.
- **Advantages**:
  - **Performance**: Enhanced performance for 64-bit applications, enabling efficient utilization of system resources.
  - **Compatibility**: Broad support for modern software packages and libraries.
  - **Security**: Regular updates and security patches ensure a secure operating environment.

1.2.2 Programming Language and Libraries

- **Python**:
  - **Version**: Python **3.9** or later is used as the primary programming language, chosen for its simplicity, versatility, and extensive ecosystem.
- **PyQt 5**:
  - Provides a comprehensive framework for developing the application's GUI, allowing the creation of responsive and feature-rich interfaces.
- **pySerial 3.5**:
  - Facilitates serial communication between the Raspberry Pi and the custom Kiosk Motor Control PCB, ensuring reliable and efficient data exchange for motor operations.
- **Peewee 3.17.6**:
  - A lightweight Object-Relational Mapping (ORM) library used to interact with the **PostgreSQL** database, simplifying database operations through an intuitive Pythonic interface.
- **Paho-MQTT 2.1.0**:
  - Implements MQTT protocol support, enabling efficient and lightweight messaging between networked devices, suitable for real-time data exchange and remote-control functionalities.
- **PostgreSQL**:
  - Serves as a robust and scalable relational database system for storing and managing application data, ensuring data integrity and efficient query performance.